

Towards a Theory of Computational Coupling Between Intelligent Systems

A Foundation for Scalable Brain-to-Brain Communication

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PART 1 — FOUNDATIONAL FIELDS

The theoretical architecture for computational coupling between intelligent systems necessitates the synthesis of distinct yet mathematically overlapping domains. Bridging biological neural networks and multi-agent reinforcement learning requires translating high-dimensional neural manifolds into low-dimensional communicative protocols, constrained by the physical limits of non-invasive hardware and the thermodynamic imperatives of living systems.

1. Information Theory and Signal Transmission

Information theory provides the strict mathematical scaffolding required to quantify the limits of communication between any two systems, independent of their physical substrate. The field was revolutionized by Claude Shannon in 1948, whose formulation of channel capacity established the absolute boundary for error-free data transmission over a noisy medium. While Shannon's original work focused on discrete and stationary channels, biological brains operate in continuous, highly non-stationary, and noisy environments.

Pioneers such as James Massey extended this work by formalizing **Directed Information** to account for systems with feedback, while Thomas Schreiber introduced **Transfer Entropy** to measure asymmetric information flow between coupled time series without requiring a model of the underlying dynamics. In the current state-of-the-art, measures like **Directed Phase Transfer Entropy (dPTE)** are actively used to map causal information flow across brain regions, distinguishing genuine communication from mere correlational synchrony.

* **Primary Limitation:** Classical information theory relies on large datasets to estimate probability distributions accurately, which struggles against the curse of dimensionality in high-density recordings.

* **Role in Computational Coupling:** Rate-Distortion Theory dictates how neural representations must be compressed before transmission, while Transfer Entropy provides the mathematical proof that one brain is actively driving the state of another, establishing causality in brain-to-brain interfaces.

2. Computational Neuroscience and Dynamical Systems

Computational neuroscience seeks to uncover the algorithmic and mechanistic principles underlying neural activity. Historically emerging from the biophysical models of Hodgkin and Huxley, the field underwent a paradigm shift through the work of David Marr, Peter Dayan, and Larry Abbott, moving toward the understanding of neural coding. The current state-of-the-art focuses heavily on **Neural Population Dynamics**. Modern techniques, analyzing data from high-density arrays like Neuropixels, demonstrate that complex behaviors are governed by low-dimensional manifolds within the high-dimensional state space of neuronal firing rates. Theories of **Coupled Dynamical Systems** and **Kuramoto Models** (pioneered by Yoshiki Kuramoto) are applied to explain how distinct populations of non-linear oscillators achieve phase-locking and generalized synchrony.

* **Primary Limitation:** Severe anatomical idiosyncrasies across subjects make it difficult to find a universal coordinate system for these manifolds.

* **Role in Computational Coupling:** Understanding neural population dynamics is required to decode the "Sender's" intent. To couple two systems, the trajectory of the Sender's neural state space must be mapped, translated, and injected into the Receiver's state space such that the receiver's dynamical system absorbs and integrates the exogenous control signals without catastrophic interference.

3. Control Theory, Consensus, and Distributed Systems

Control theory provides the formalisms for driving a dynamical system from an initial state to a desired target state in the presence of noise and transmission delays. Originating from Norbert Wiener's cybernetics and Rudolf Kalman's

state estimation filters, modern applications have expanded into **Networked Control Systems** and **Distributed Systems**. Within these networks, **Consensus Algorithms** govern how independent nodes agree on a shared state, forming the basis of Collective Intelligence in complex systems.

The current state-of-the-art utilizes optimal control and model predictive control to manage multi-agent swarms and stabilize delayed non-linear networks.

* **Primary Limitation:** Unobservability of the full brain state and the highly non-linear, adaptive nature of neural plasticity.

* **Role in Computational Coupling:** Computational coupling between brains is fundamentally a networked optimal control problem. The Sender brain acts as the controller, attempting to drive the Receiver brain (the plant) into a specific representational state across a narrow-bandwidth channel, requiring sophisticated consensus protocols to align their internal states.

4. Active Inference and Predictive Coding

Active inference provides a unifying neurobiological theory based on the premise that all sentient systems minimize variational free energy to resist the second law of thermodynamics. Championed by Karl Friston, this framework posits that the brain is a statistical prediction engine that updates its internal generative models to explain away sensory prediction errors (perception) or acts upon the world to fulfill its predictions (action).

Recently, the Free Energy Principle has been extended to social interactions and communication. When two agents interact, they form a coupled system where each attempts to predict the other. Communication emerges as a mechanism for "mental alignment," where agents minimize joint surprise to establish a shared narrative or hermeneutic niche. State-of-the-art models currently simulate active inference using **Partially Observable Markov Decision Processes (POMDPs)** to explain parent-infant synchrony and therapeutic alliances.

* **Primary Limitation:** Computing exact free energy is intractable for open-ended, high-dimensional human interactions.

* **Role in Computational Coupling:** Active inference offers the exact teleological justification for why two connected brains would synchronize: coupling minimizes the variational free energy of the joint system, driving the emergence of a shared inter-brain communication protocol.

5. Representation Learning and Brain Foundation Models

Representation learning leverages deep neural networks to extract low-dimensional, semantically rich features from raw, unstructured data. Building upon self-supervised learning (SSL) breakthroughs in natural language processing (e.g., Transformers), researchers have recently pioneered **Brain Foundation Models**. Groups led by David van Dijk, Bao-Liang Lu, and researchers at Meta AI have introduced massive models pre-trained on tens of thousands of hours of neuroimaging data.

Current SOTA models include **BrainLM** (trained on 6,700 hours of fMRI) and **LaBraM/LUNA** (trained on massive clinical EEG corpora using vector-quantized neural spectrum prediction). These models demonstrate powerful zero-shot inference, decoding cognitive states and extracting universal neural tokens across subjects.

* **Primary Limitation:** Recording modalities constrain capabilities: fMRI-based models suffer from extreme hemodynamic latency (seconds), while EEG models lack precise spatial resolution.

* **Role in Computational Coupling:** Scalable brain-to-brain communication relies entirely on these foundation models. They serve as the universal "interlingua" or codec, translating the idiosyncratic neural patterns of the Sender into a subject-agnostic latent space, which is then re-personalized for the Receiver.

6. Brain-Computer and Brain-to-Brain Interfaces

The empirical pursuit of directly linking biological nervous systems to digital machines or to each other.

Brain-Computer Interfaces (BCIs) originated with early single-unit recordings in primates, evolving toward human motor prostheses. Brain-to-Brain Interfaces (BBIs) emerged when Miguel Nicolelis demonstrated the first rat-to-rat sensorimotor transfer via intracortical microstimulation (ICMS) in 2013.

Shortly after, Rajesh Rao and Andrea Stocco demonstrated the first non-invasive human BBI using EEG and Transcranial Magnetic Stimulation (TMS), later expanding this to a three-person network (**BrainNet**) in 2019. Current SOTA human BBIs allow collaborative problem-solving (e.g., Tetris) by decoding motor imagery and injecting binary information via TMS phosphenes into the visual cortex.

* **Primary Limitation:** Severe bandwidth bottleneck in non-invasive hardware, currently restricted to ~1 bit per second, forcing communication to remain rudimentary and discrete.

* **Role in Computational Coupling:** Moving beyond binary phosphenes to high-bandwidth semantic computational coupling requires bridging the hardware constraints of BBI with the software representations of foundation models.

7. Optimal Transport and Functional Alignment

Because every human brain differs macroscopically (gyral folding) and microscopically (functional tuning), transferring information directly from spatial coordinates yields catastrophic misalignment. **Hyperalignment**, introduced by James Haxby, first solved this by using Procrustes transformations to align data in a high-dimensional information space rather than anatomical space. The **Shared Response Model (SRM)** advanced this through probabilistic generative factorizations.

The current mathematical SOTA is **Fused Unbalanced Gromov-Wasserstein (FUGW)** alignment, pioneered by Alexis Thual and Bertrand Thirion. FUGW utilizes Optimal Transport to match cortical surfaces based on functional similarity (Wasserstein distance) while penalizing severe topological distortions (Gromov-Wasserstein distance), permitting unequal mass distributions across subjects.

* **Primary Limitation:** Intense computational cost of solving non-convex block-coordinate descent problems over dense meshes.

* **Role in Computational Coupling:** FUGW provides the exact mathematical routing protocol required. It ensures that a semantic vector extracted from the Sender's visual cortex is injected into the topologically and functionally homologous neural population in the Receiver.

8. Multi-Agent Reinforcement Learning (MARL) and Emergent Communication

MARL explores how independent agents acting in a shared environment learn policies to maximize global or local rewards. A critical subfield is **Emergent Communication**, where agents invent novel protocols to solve Decentralized Partially Observable Markov Decision Processes (Dec-POMDPs).

Foerster and Whiteson pioneered this with **Differentiable Inter-Agent Learning (DIAL)**, showing that deep neural networks can backpropagate gradients through noisy communication channels, allowing continuous and discrete languages to emerge entirely from optimization. The SOTA involves using contrastive learning to force messages to capture global state representations, achieving zero-shot coordination.

* **Primary Limitation:** Non-stationarity (the moving target problem as all agents learn simultaneously) and a failure to generalize to unseen partners.

* **Role in Computational Coupling:** MARL provides the artificial analog to BBI. When two brains are coupled via a digital channel, they form a MARL system. The theory of computational coupling must leverage DIAL-like architectures to facilitate the emergence of a highly compressed, bandwidth-efficient "neural language" between the biological subjects.

9. Neuroscience of Communication and Hyperscanning

The neurobiological study of how brains synchronize during naturalistic interaction. Read Montague pioneered **Hyperscanning** in 2002 by linking two fMRI scanners over the internet to study the neural substrates of deception. Today, SOTA research utilizes EEG and fNIRS hyperscanning in highly ecological settings to measure **Inter-Brain Synchrony (IBS)** during joint attention, collaboration, and dialogue. These studies consistently show phase-locking in prefrontal and temporoparietal networks during successful cooperation.

* **Primary Limitation:** Proving causality: much of the observed neural synchrony is an epiphenomenon of two brains receiving identical sensory stimuli (e.g., hearing the same sound) rather than true brain-to-brain coupling.

* **Role in Computational Coupling:** Hyperscanning provides the baseline observational data for your research. A true theory of computational coupling must transcend observational hyperscanning by artificially inducing these synchronized states through targeted neural encoding.

PART 2 — COMPLETE PAPER DATABASE

| Title | Authors | Year | Venue | DOI / Link | Abstract Summary | Main Contribution | Math/Exp Contribution | Relation to Work | Sim Score / Threat | Weaknesses | Future Work |

| :-- | :-- | :-- | :-- | :-- | :-- | :-- | :-- | :-- | :-- | :-- |

| **A Brain-to-Brain Interface for Real-Time Sharing of Sensorimotor Information** | Pais-Vieira, Lebedev, Nicolelis | 2013 | Sci Rep | `10.1038/srep01319` | Real-time transfer of sensorimotor info between rats. Encoder rat performed tasks, activity transmitted to decoder via ICMS. | First empirical demonstration of a BBI between mammals. | ~70% behavioral transfer accuracy via linear decoding & microstimulation. | Establishes biological plausibility of direct computational coupling. | **70% (Low)** | Invasive; limited to basic binary discrimination tasks. | Scale to complex biological computing networks. |

| **A Direct Brain-to-Brain Interface in Humans** | Rao, Stocco, Bryan, Prat | 2014 | PLoS ONE | `10.1371/journal.pone.0111332` | First non-invasive human BBI. Uses EEG motor imagery in sender to trigger TMS in receiver. | Proves human BBI is achievable safely without surgery. | End-to-end latency and accuracy quantification across the internet. | Foundational baseline for non-invasive BBI frameworks. | **75% (Low)** | Unidirectional; extremely low bit-rate; relies on motor output. | Bidirectional closed-loop systems. |

| **Learning to Communicate with Deep Multi-Agent RL** | Foerster, Assael, de Freitas | 2016 | NIPS | `arxiv:1605.06676` | Agents learn communication protocols to solve tasks by backpropagating through noisy channels. | Introduces DIAL (Differentiable Inter-Agent Learning). | Centralized learning with decentralized execution for communication. | Exact mathematical framework for emergent BBI protocol. | **75% (Medium)** | Tested only on discrete grid-world proxies. | Generalization to continuous biological systems. |

| **BrainNet: A Multi-Person Brain-to-Brain Interface...** | Jiang, Stocco, Rao | 2019 | Sci Rep | `10.1038/s41598-019-41895-7` | 3-person collaborative BBI playing Tetris. Receivers learn to trust reliable senders via injected noise. | First multi-node human BBI (Brainet); demonstrates social trust emerging via neural link. | Calculation of Mutual Information to estimate signal reliability. | Moves BBI from dyadic pairs to scalable multi-agent networks. | **85% (Medium)** | Bandwidth strictly limited to binary phosphene triggers. | Scaling to semantic data structures. |

| **Aligning individual brains with Fused Unbalanced Gromov-Wasserstein** | Thual, Tran, Thirion | 2022 | NeurIPS | `arxiv:2206.09398` | Inter-subject alignment using Optimal Transport to match functional signatures while penalizing anatomical deformation. | Introduces FUGW solver for whole-brain, landmark-free alignment. | Interpolates between Wasserstein (functional) and Gromov-Wasserstein (geometric) losses. | Mathematical routing protocol to map high-dim data from Sender to Receiver. | **70% (Low)** | High computational cost due to non-convex block-coordinate descent. | Faster approximation solvers. |

| **Therapeutic Alliance as Active Inference** | McParlin, Cerritelli, Friston | 2022 | Front Behav Neurosci | `10.3389/fnbeh.2022.897247` | Empirical account of biobehavioral synchrony and communication using active inference. | Formalizes interpersonal alignment as mutual free energy minimization. | Establishes generative models for coupled sentient systems. | Theoretical "why" for computational coupling alignment. | **85% (High)** | Highly theoretical; lacks direct neural intervention data. | Empirical validation of joint POMDPs. |

| **Large Brain Model for Learning Generic Representations (LaBraM)** | Jiang, Zhao, Lu | 2024 | ICLR | `arxiv:2405.18765` | EEG foundation model trained on 2,500 hours of data using a vector-quantized neural tokenizer. | Achieves SOTA across multiple EEG tasks via unsupervised pre-training. | Implements Vector-Quantized Neural Spectrum Prediction (VQ-NSP). | Latent space acts as generic "codec" for coupling EEGs. | **80% (Medium)** | Focuses solely on offline decoding; ignores encoding/writing. | Real-time zero-shot inference. |

| **BrainLM: A foundation model for brain activity recordings** | Caro, Fonseca, van Dijk | 2024 | bioRxiv | `10.1101/2023.09.12.557460` | Generative foundation model trained on 6,700 hours of fMRI to capture spatiotemporal dynamics. | Identifies intrinsic functional networks in zero-shot mode without supervision. | Masked autoencoding across 424 AAL brain parcels. | fMRI baseline for extracting rich, high-level semantic states. | **70% (Low)** | fMRI temporal lag severely limits applicability to real-time interaction. | Multimodal fusion with EEG. |

| **MindEye2: Shared-Subject Models Enable fMRI-To-Image...** | Scotti et al. | 2024 | ICML | `arxiv:2403.11207` | Reconstructs visual perception using shared-subject alignment mapped to CLIP latent space. | Solves data-scarcity problem for cross-subject neural decoding. | Ridge regression mapping to shared unCLIP space. | Proves high-bandwidth semantic extraction is feasible cross-subject. | **90% (High)** | Only works for visual stimuli; susceptible to noise/motion. | Generalizing beyond visual cortex to abstract thoughts. |

| **Unsupervised method for representation transfer from one brain to another** | Nakamura, Kanai, Hayashi | 2024 | Front Neuroinf | `10.3389/fninf.2024.1470845` | Transforms neural representations between subjects without corresponding label information. | Solves zero-shot alignment problem via unsupervised geometric embedding. | Instance learning rules mapping to n-dimensional hyperspheres. | Threatens novelty of representation translation layer. | **95% (High Threat)** | Requires pre-defined anatomical masks; limited empirical tasks. | Full-brain unsupervised alignment. |

| **A Reduced-Dimension fMRI Shared Response Model** | Chen, Haxby, Ramadge | 2015 | NeurIPS | `10.5555/2969442.2969507` | Aggregates multi-subject fMRI by mapping to a shared response vector matrix. | Introduces SRM for functional alignment based on time-locked stimuli. | Factorizes $X_i = W_i S + E_i$ subject to orthogonality constraints. | Historical baseline for creating a shared latent representation across brains. | **60% (Low)** | Requires exact time-locked stimuli across subjects. | Asynchronous alignment models. |

| **Hyperscanning: Simultaneous fMRI during linked social interactions** | Montague et al. | 2002 | NeuroImage | `10.1006/nimg.2002.1150` | Links two fMRI scanners via internet to study deception games. | Invents hyperscanning as a field. | Establishes cross-correlation metrics for dual-scanner data. | Baseline methodology for observing naturalistic inter-brain synchrony. | **50% (Low)** | Purely observational; massive latency issues. | Incorporating real-time neurofeedback. |

PART 3 — HISTORICAL TIMELINE

1948 — Information Theory is Born: Claude Shannon publishes *A Mathematical Theory of Communication**, defining entropy and establishing that all channels have a calculable capacity for error-free transmission.

* **1969 — The Formulation of Causality:** Clive Granger formalizes causality in time series, creating the statistical foundation for proving directional influence between variables.

* **1975 — Synchronization of Oscillators:** Yoshiki Kuramoto formulates the Kuramoto model, mathematically proving how populations of independent, non-linear oscillators spontaneously phase-lock.

* **1990 — Directed Information:** James Massey extends Shannon's work, developing Directed Information to properly model communication over channels that contain feedback loops.

* **2000 — Transfer Entropy:** Thomas Schreiber introduces Transfer Entropy, a model-free, information-theoretic measure to quantify asymmetric information flow between dynamical systems.

* **2002 — The Birth of Hyperscanning:** Read Montague conducts the first simultaneous fMRI experiment, connecting scanners over the internet to observe inter-brain dynamics during social deception games.

* **2010 — The Free Energy Principle:** Karl Friston consolidates predictive coding into a unified brain theory. The brain is modeled as a Bayesian engine minimizing variational free energy (surprise).

* **2011 — Hyperalignment:** James Haxby solves the anatomical variability problem in fMRI by aligning functional representations in a high-dimensional Procrustes space.

* **2013 — The First Brain-to-Brain Interface:** Miguel Nicolelis's lab demonstrates the first BTBI, enabling real-time transfer of sensorimotor decisions between the cortices of two rats via microstimulation.

* **2014 — Non-Invasive Human BBI:** Rajesh Rao and Andrea Stocco construct the first non-invasive human BBI, allowing one human to control the motor actions of another via EEG and internet-transmitted TMS.

* **2015 — Shared Response Models & Active Inference for Communication:** Chen et al. refine alignment with probabilistic SRMs. Simultaneously, Friston and Frith apply active inference to communication, showing that brains synchronize to minimize mutual prediction errors.

* **2016 — Emergent AI Communication:** Foerster et al. introduce DIAL, proving that deep neural networks acting as independent agents can invent proprietary communication protocols to solve cooperative tasks when gradients are backpropagated across the channel.

* **2019 — Multi-Agent Brainets:** Jiang, Stocco, and Rao expand dyadic BBIs into "BrainNet," connecting three human brains to collaboratively play Tetris.

* **2022 — Optimal Transport for Brain Alignment:** Thual et al. introduce FUGW, utilizing optimal transport mathematics to achieve landmark-free, whole-brain functional alignment while preserving cortical geometry.

* **2024 — Brain Foundation Models & Zero-Shot Alignment:** Explosion of self-supervised learning yields LaBraM, BrainLM, and MindEye2. Nakamura et al. demonstrate unsupervised representation transfer across brains using hyper-spherical embeddings.

PART 4 — RESEARCH GROUPS

| Organization | Principal Investigators | Major Projects / Repositories | Current Research Direction |

| :--- | :--- | :--- | :--- |

| **Center for Neurotechnology (UW)** | Rajesh Rao, Andrea Stocco, Chantel Prat | BrainNet, Neural Co-processors | Human BBIs, predictive coding, cooperative problem solving over neural networks. |

| **Nicolelis Lab (Duke University)** | Miguel Nicolelis | Walk Again Project, Animal Brainets | Highly invasive BMIs, multi-agent animal computing clusters via ICMS. |

| **Inria / NeuroSpin (Parietal)** | Bertrand Thirion, Alexis Thual, Rémi Flamary | `fugw` (FUGW solver), `Nilearn` | Optimal transport for brain alignment, solving non-convex optimization for dense cortical meshes. |

| **Wellcome Centre (UCL)** | Karl Friston, Christopher Frith | `SPM` (Statistical Parametric Mapping) | Active inference, free energy principle, computational psychiatry, hermeneutics. |

| **Araya Inc. / RIKEN (Japan)** | Ryota Kanai, Ryusuke Hayashi, M. Oizumi | `NeuRep\_GWOT` | Artificial general intelligence, consciousness, unsupervised representation transfer across brains. |

| **SJTU (Dongsheng Li Group)** | Bao-Liang Lu, Wei-Bang Jiang | LaBraM, NeuroLM | Large Brain Models, EEG foundation models via vector quantization, emotion recognition. |

| **Van Dijk Lab (Yale)** | David van Dijk | BrainLM | Generative fMRI foundation models, self-supervised masking for clinical biomarker prediction. |

| **Meta AI (FAIR)** | Jean-Rémi King | `brainmagick`, MEG-to-Speech | Decoding speech and semantics from non-invasive MEG using self-supervised contrastive learning. |

| **Neuralink** | N/A (Corporate) | N1 Implant, R1 Robot | Invasive high-density (1024 channel) thread implantation for high-bandwidth read/write access. |

| **Synchron** | Thomas Oxley | Stentrode | Endovascular brain-computer interfaces requiring no open brain surgery for locked-in patients. |

| **Precision Neuroscience** | Benjamin Rapoport | Layer 7 Cortical Interface | High-density surface micro-electrode arrays placed via minimally invasive cranial micro-slits. |

| **Paradromics** | Matt Angle | Connexus Direct Data Interface | High data-rate invasive cortical implants focusing on speech decoding. |

| **Google DeepMind** | N/A (Corporate) | Multi-Agent RL frameworks | Zero-shot coordination, emergent communication, collective intelligence modeling. |

| **Princeton Neuroscience** | Uri Hasson, Peter Ramadge | BrainIAK, Shared Response Model | Inter-subject correlation, naturalistic fMRI stimuli, hyperalignment. |

PART 5 — DATASETS

| Dataset Name | Modality | Subjects | Tasks / Labels | License | Recommended Use | Advantages & Limitations |

| :--- | :--- | :--- | :--- | :--- | :--- | :--- |

| **Natural Scenes Dataset (NSD)** | 7T fMRI | 8 | 73,000 COCO images; densely sampled | ODC-By | Foundation models (MindEye2), neural decoding. | **Adv:** Unprecedented depth per subject. **Lim:** Only 8 subjects; highly visual focus. |

| **UK Biobank (Neuroimaging)** | fMRI | >40,000 | Rest, emotion, clinical metadata | Restricted | Pre-training large-scale structural models (BrainLM). | **Adv:** Massive scale, rich clinical data. **Lim:** Complex access requirements; low temporal resolution (0.73s). |

| **Human Connectome Project (HCP)** | fMRI / MEG | >1,000 | High-res structural, functional, behavioral | Open | Cross-modal functional alignment, connectivity baseline. | **Adv:** Extremely clean, high-quality data. **Lim:** Mostly

healthy, resting-state bias. |

| **TUH EEG Corpus** | Clinical EEG | >10,000 | Abnormalities, seizures, slowing | Research | Pre-training EEG models (LaBraM/LUNA), artifact detection. | **Adv:** Largest open EEG corpus; high ecological validity. **Lim:** Massive noise, varied electrode montages. |

| **FOMO300K** | 3D MRI | 59,969 | Structural scans, anomalies, tumors | CC-BY | SSL for medical vision, geometric anatomy alignment. | **Adv:** Highest heterogeneity and scale. **Lim:** Structural only; no temporal dynamics. |

| **Narrative Movie fMRI ('ds005531')** | fMRI | 69 | 69 hours of movies, LLM annotations | CC0 | Inter-brain synchrony, shared response modeling. | **Adv:** Deep linguistic/semantic annotations. **Lim:** Tiny subject pool. |

| **SEED-V** | EEG | 16 | Emotion recognition via video clips | Research | Fine-tuning downstream affective BCIs. | **Adv:** High-quality affective labels. **Lim:** Lab-constrained artificial emotions. |

PART 6 — BENCHMARKS

| Benchmark | Relevance / Domain | What it Measures | Why it Exists | Current SOTA |

| :--- | :--- | :--- | :--- | :--- |

| **MOABB (Mother of All BCI Benchmarks)** | Neural Decoding | Classification accuracy across Motor Imagery, P300, and SSVEP tasks on varied datasets. | BCI literature suffers from a reproducibility crisis with highly customized, non-comparable pipelines. | BrainDecode models; Deep Learning hybrids. |

| **LUNA / BioFoundation Benchmarks** | Brain Foundation Models | AUROC and Balanced Accuracy on TUAB (Abnormality) and TUAR (Artifact) datasets. | Assesses zero-shot and transfer-learning capability of large-scale EEG foundation models. | LUNA (Topology-Agnostic Transformer) & FEMBA. |

| **MindEye fMRI-to-Image Benchmarks** | Cross-Subject Learning | PixCorr, SSIM, Inception distance, and CLIP similarity between predicted and ground-truth images. | Evaluates whether neural representations of semantics can generalize across heterogeneous brains. | MindEye2 (using shared-subject ridge regression). |

| **Zero-Shot Communication (MARL)** | Multi-Agent Communication | Reward achieved by an agent when paired with a novel partner not seen during training. | Tests if emergent communication protocols are generalized "languages" or overfit cryptographic handshakes. | CACL (Contrastive Action-Communication Learning). |

PART 7 — MATHEMATICAL TOOLS

1. Information-Theoretic Causality

Transfer Entropy (TE)

* **Intuition:** Measures how much uncertainty about the future of signal Y_t is reduced by knowing the past of signal X_t , given the past of Y_t . It proves asymmetric, directional information flow.

* **Formal Definition:**

$$T_{X \rightarrow Y} = H(Y_t | Y_{t-1:t-L}) - H(Y_t | Y_{t-1:t-L}, X_{t-1:t-L})$$

* **Applications:** Distinguishing true computational coupling from superficial hyperscanning synchrony.

* **Key Paper:** Schreiber (2000).

Directed Information

* **Intuition:** An extension of mutual information for systems with feedback, summing conditional mutual information over time.

* **Formal Definition:**

$$I(X^N \rightarrow Y^N) = \sum_{i=1}^N I(X^i; Y_i | Y^{i-1})$$

* **Key Paper:** Massey (1990).

Rate-Distortion Theory

* **Intuition:** Calculates the minimum channel capacity (rate) required to transmit a signal while keeping the reconstruction error (distortion) below a specific threshold. Vital for compressing neural state spaces to fit through narrow BBI channels.

2. Optimal Transport & Differential Geometry

Fused Unbalanced Gromov-Wasserstein (FUGW) Alignment

* **Intuition:** Classic optimal transport (Wasserstein) matches points in a shared space. Gromov-Wasserstein matches metric spaces based on internal geometric distances, bypassing the need for a shared coordinate system. "Fused" combines both. "Unbalanced" allows mass creation/destruction, solving the problem of brains having differently sized functional regions.

* **Formal Loss Function:**

$$\mathcal{L}_{\{\alpha, \rho, \epsilon\}}(P) = (1-\alpha) \sum_{i,j} \|F^s_i - F^t_j\|^2 P_{i,j} + \alpha \sum_{i,j,k,l} |D^s_{i,k} - D^t_{j,l}|^2 P_{i,j} P_{k,l} + \rho_{\{KL\}}(P) + \epsilon \mathbb{E}(P)$$

* **Applications:** Routing semantic neural representations from the Sender's topology to the homologous region in the Receiver.

* **Key Paper:** Thual et al. (2022).

3. Bayesian Mechanics & Dynamical Systems

Variational Free Energy (Active Inference)

* **Intuition:** Systems bound surprise (negative log marginal likelihood) by minimizing free energy. Perception updates the internal model; action changes sensory data. Coupling occurs when two systems act to minimize joint prediction errors.

* **Formal Definition:**

$$F = \mathbb{E}_{q(s)}[\log q(s) - \log p(o, s)] = D_{\{KL\}}[q(s) \parallel p(s)] - \mathbb{E}_{q(s)}[\log p(o \mid s)]$$

* **Key Papers:** Friston (2010), Friston & Frith (2015).

Lyapunov Stability & Koopman Operators

* **Intuition:** Lyapunov stability proves whether a coupled dynamical system will converge to a synchronized equilibrium. Koopman operator theory allows non-linear finite-dimensional dynamical systems to be represented as linear infinite-dimensional systems, making complex neural population dynamics tractable for control theory.

Neural ODEs & State Space Models (SSMs)

* **Intuition:** Modeling neural dynamics as continuous-time differential equations (Neural ODEs) or continuous-discrete mapping systems (Mamba/SSMs). Critical for forecasting the trajectory of the Receiver brain upon signal injection.

4. Reinforcement Learning & Stochastic Processes

Markov Decision Processes (MDP) & POMDPs

* **Intuition:** Framework for decision making under uncertainty. Active inference models communication as a POMDP where the other agent's internal state is the hidden variable s_s to be inferred from observations o_s .

Information Bottleneck

* **Intuition:** Extracts relevant information from X about Y by compressing X into a latent variable Z , maximizing $I(Z; Y)$ while minimizing $I(X; Z)$. Central to modern emergent communication algorithms (e.g., DIAL).

PART 8 — BOOKS & CORE TEXTS

1. Cover, T. M., & Thomas, J. A. — *Elements of Information Theory*

Why it matters: The undeniable bedrock for understanding entropy, channel capacity, and rate-distortion. You cannot formally define bandwidth constraints without it.

2. Parr, T., Pezzulo, G., & Friston, K. J. — *Active Inference: The Free Energy Principle in Mind, Brain, and Behavior*

Why it matters: Translates Karl Friston's dense mathematics into comprehensible POMDP models, essential for modeling communication as mutual free-energy minimization.

3. Dayan, P., & Abbott, L. F. — *Theoretical Neuroscience: Computational and Mathematical Modeling of Neural Systems*

Why it matters: The canonical text bridging neurobiology with computational models (encoding, decoding, network dynamics).

4. **Villani, C.** — *Optimal Transport: Old and New*

Why it matters: The definitive mathematical treatise on Wasserstein distances. Required reading to deeply understand and modify FUGW alignment algorithms.

5. **Sutton, R. S., & Barto, A. G.** — *Reinforcement Learning: An Introduction*

Why it matters: The baseline for understanding Multi-Agent Reinforcement Learning and the mathematical foundations of emergent communication (DIAL/CACL).

PART 9 — NOVELTY ANALYSIS & ADVERSARIAL DEFENSE

1. Dangerously Close Papers & Novelty Threats

* **Nakamura et al. (2024)** — **Unsupervised method for representation transfer from one brain to another:**

Threat: Directly threatens novelty regarding zero-shot cross-subject alignment using instance learning on n-dimensional hyperspheres.

Defense: Our theory progresses beyond passive representation transfer into active, closed-loop control (driving the receiver's neural state trajectory via feedback).

* **Scotti et al. (2024)** — **MindEye2:**

Threat: Solved the data-scarcity problem for fMRI semantic extraction via shared-subject ridge regression mapping to CLIP space.

Defense: Our theory does not merely align static semantic spaces; it solves the continuous temporal, causal, and dynamic injection problem over bandwidth-limited channels.

2. Core Concepts That Could Invalidate Work

* **The "Superficial Synchrony" Illusion:** Hyperscanning literature is plagued by the artifact of shared sensory input. If two brains watch the same movie, they synchronize, but they are not computationally coupled.

Defense: We deploy Directed Information and Transfer Entropy to mathematically separate stimulus-driven correlation from true causal injection.

* **The Hemodynamic Lag Trap:** fMRI foundation models (BrainLM, MindEye) have a 4-6 second BOLD latency.

Continuous MARL communication requires millisecond precision.

Defense: Synthesize fMRI spatial mapping (FUGW) with EEG temporal dynamics (LaBraM) to form a unified spatio-temporal model.

3. Assumptions We Must Prove

* **Manifold Isomorphism:** We assume underlying neural manifolds of different subjects are sufficiently isomorphic to allow non-destructive topological transfer.

* **Hardware Agnosticism:** We must prove that our mathematical theory holds regardless of whether the physical channel is a 1-bit/sec TMS pulse or a 1024-channel ICMS array.

PART 10 — FINAL OUTPUT & EXECUTION PLAN

1. Top 15 Most Important Papers (The Core Canon)

1. **Foerster et al. (2016)** — *Learning to Communicate with Deep MARL* (AI mechanism)

2. **Thual et al. (2022)** — *Aligning individual brains with FUGW* (Mapping mechanism)

3. **Jiang et al. (2019)** — *BrainNet* (BBI empirical baseline)

4. **Friston & Frith (2015)** — *Active inference, communication and hermeneutics* (Biological teleology)

5. **Nakamura et al. (2024)** — *Unsupervised method for representation transfer* (Primary novelty threat)

6. **Scotti et al. (2024)** — *MindEye2* (SOTA semantic alignment)

7. **Jiang et al. (2024)** — *LaBraM* (SOTA temporal representation)

8. **Caro et al. (2024)** — *BrainLM* (SOTA spatial representation)

9. **Schreiber (2000)** — *Measuring Information Transfer* (Causality metric)

10. **Chen et al. (2015)** — *Shared Response Model* (Alignment baseline)
11. **Hasson et al. (2004)** — *Intersubject synchronization of cortical activity* (Hyperscanning foundation)
12. **Rao et al. (2014)** — *A Direct Brain-to-Brain Interface in Humans* (BBI foundation)
13. **Pais-Vieira et al. (2013)** — *A Brain-to-Brain Interface for Real-Time Sharing...* (Animal ICMS baseline)
14. **McParlin et al. (2022)** — *Therapeutic Alliance as Active Inference* (Biobehavioral synchrony)
15. **Lin et al. (2021)** — *Learning to Ground Multi-Agent Communication* (SOTA MARL communication)

2. Top 5 Open-Source Repositories

1. [alexisthual/fugw](https://github.com/alexisthual/fugw) — FUGW Optimal Transport solvers
2. [935963004/LaBraM](https://github.com/935963004/LaBraM) — EEG Foundation Model
3. [vandijklab/brainlm](https://github.com/vandijklab/brainlm) — fMRI Foundation Model
4. [pulp-bio/biofoundation](https://github.com/pulp-bio/biofoundation) — LUNA / Topology-Agnostic EEG
5. [NeuroDecode/MOABB](https://github.com/NeuroDecode/MOABB) — Mother of all BCI Benchmarks

3. Mathematical Dependency Graph

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Information Theory (Shannon, Rate-Distortion, TE)



▼ (Establishes what can be transmitted & proves causality)

Dynamical Systems & Control Theory (Kuramoto, Lyapunov)



▼ (Establishes physical & coupled behavior)

Optimal Transport (Villani, FUGW)



▼ (Establishes geometric transfer across idiosyncratic brains)

Representation Learning (BrainLM, LaBraM)



▼ (Provides latent codec for extraction)

Emergent Communication (MARL, DIAL)



▼ (Provides protocol optimization objective)

Active Inference (Friston)



▼ (Provides biological imperative for receiver alignment)

Unified Theory of Computational Coupling

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4. Prioritized 6-Month Roadmap

* **Month 1 (Mathematical Foundations):** Deep dive into Optimal Transport. Mathematically dismantle the FUGW objective function. Clone `fugw` and run synthetic manifold alignment experiments to understand entropic regularization ϵ .

* **Month 2 (Neural Encoding/Decoding):** Study BrainLM, MindEye2, and LaBraM architectures. Analyze how masked autoencoding and vector-quantized tokenizers compress neural data. Set up Natural Scenes Dataset (NSD) locally.

* **Month 3 (Active Inference & Teleology):** Formulate communication mathematically as a POMDP where minimizing variational free energy drives the Receiver to adopt the Sender's state.

* **Month 4 (MARL & Emergent Communication):** Implement Foerster's DIAL architecture. Engineer a custom loss function that mimics BBI constraints (narrow bandwidth choking, latency).

* **Month 5 (Synthesis & Formulation):** Begin drafting foundational mathematics of the paper. Construct unified theory: Sender state extraction (LaBraM/BrainLM) $\rightarrow$ Geometric alignment (FUGW) $\rightarrow$ Rate-Distortion compression $\rightarrow$ Biological integration (Active Inference).

* **Month 6 (Novelty Defense & Polish):** Conduct rigorous theoretical defense against Nakamura (2024) and hyperscanning "superficial synchrony" critiques. Refine proofs relying on Directed Information and Transfer Entropy to guarantee causal coupling.